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(54) **TAMPER-RESISTANT DOOR LOCKING SYSTEM**

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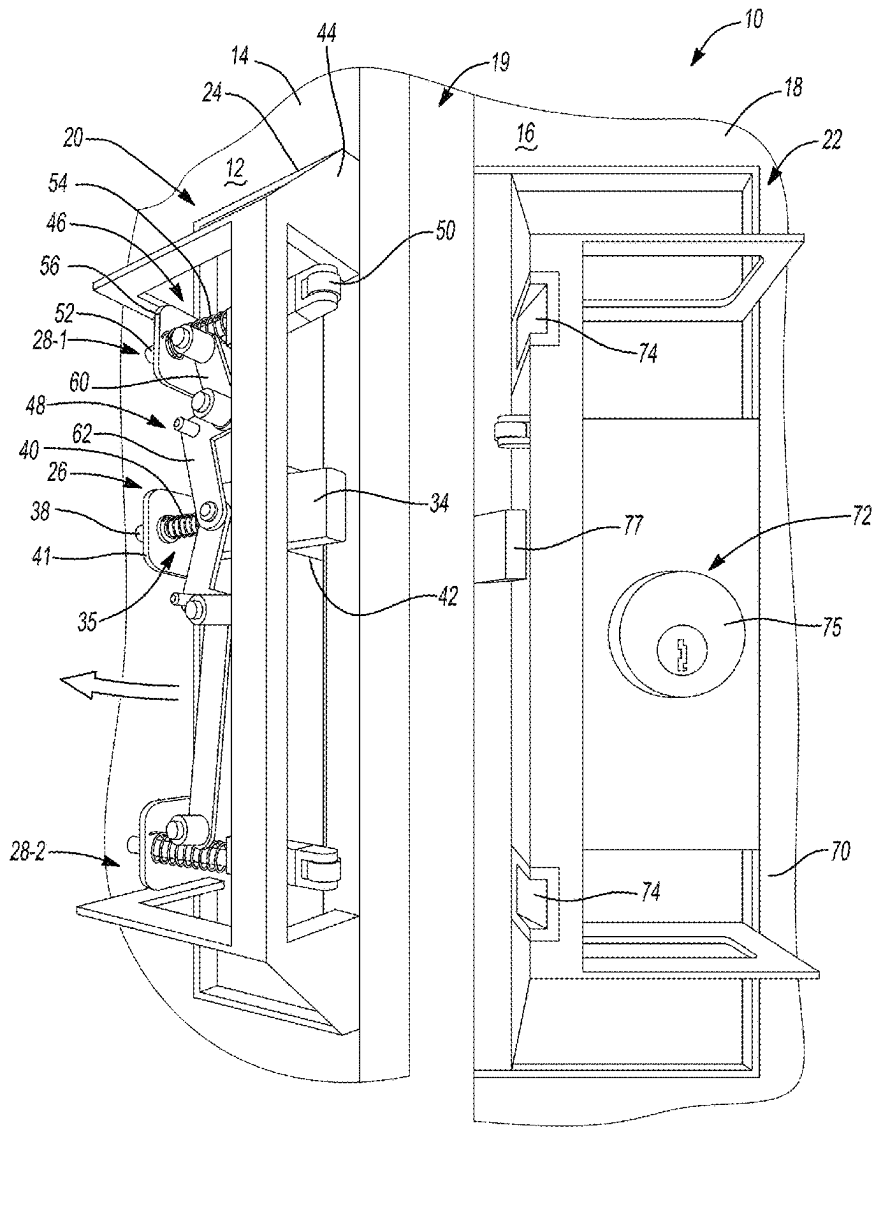
(57) **ABSTRACT**

A lock system includes a first housing, a plug, and a deadlock system. The first housing including a first side defining an opening. The plug is disposed within the first housing and is configured to move within the opening between an open position and a plugged position. The deadlock system is disposed within the first side of the first housing and coupled to the plug. The deadlock system is configured to move between a first position and a second position. The deadlock system (i) inhibits movement of the plug in the first position and (ii) allows movement of the plug in the second position.



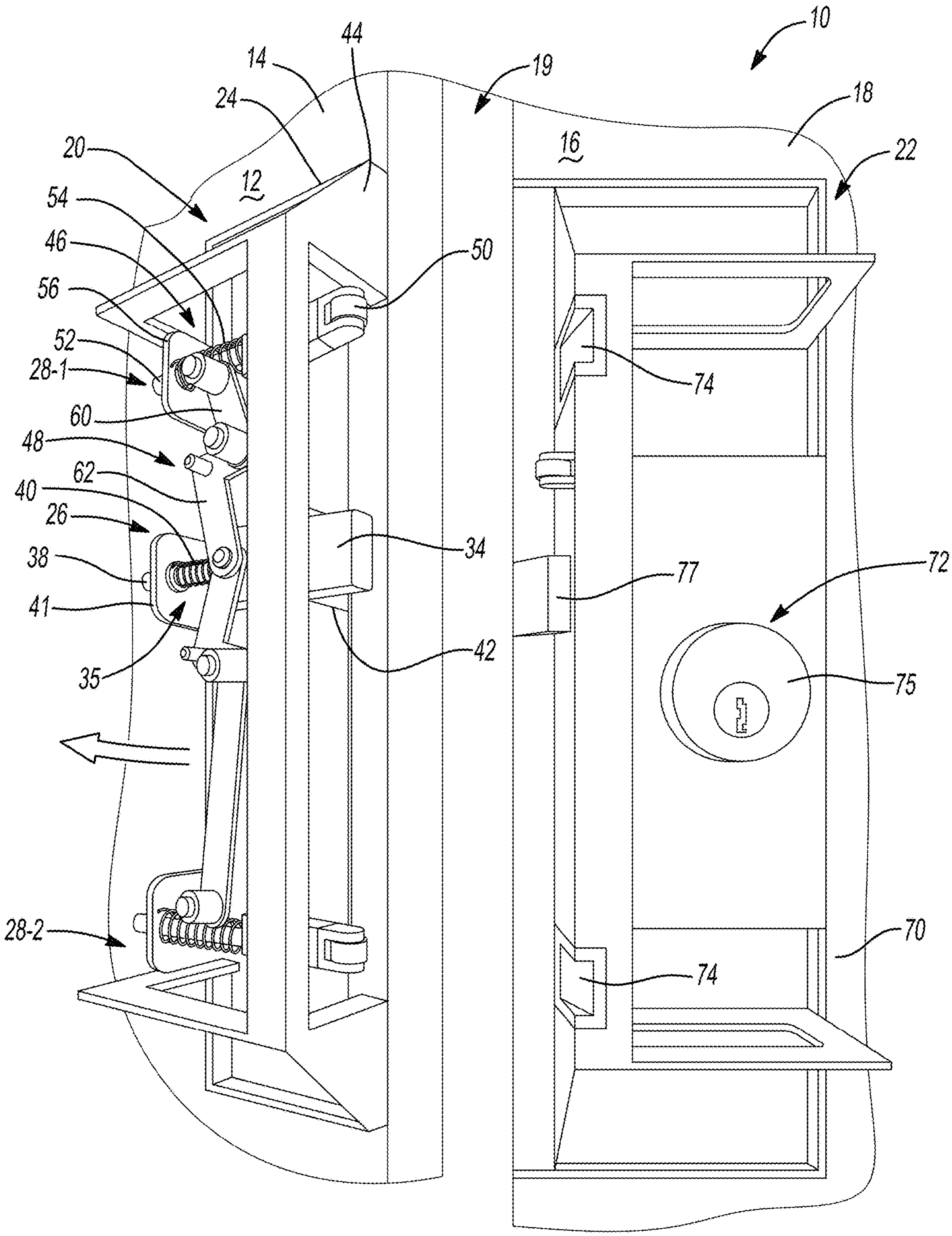

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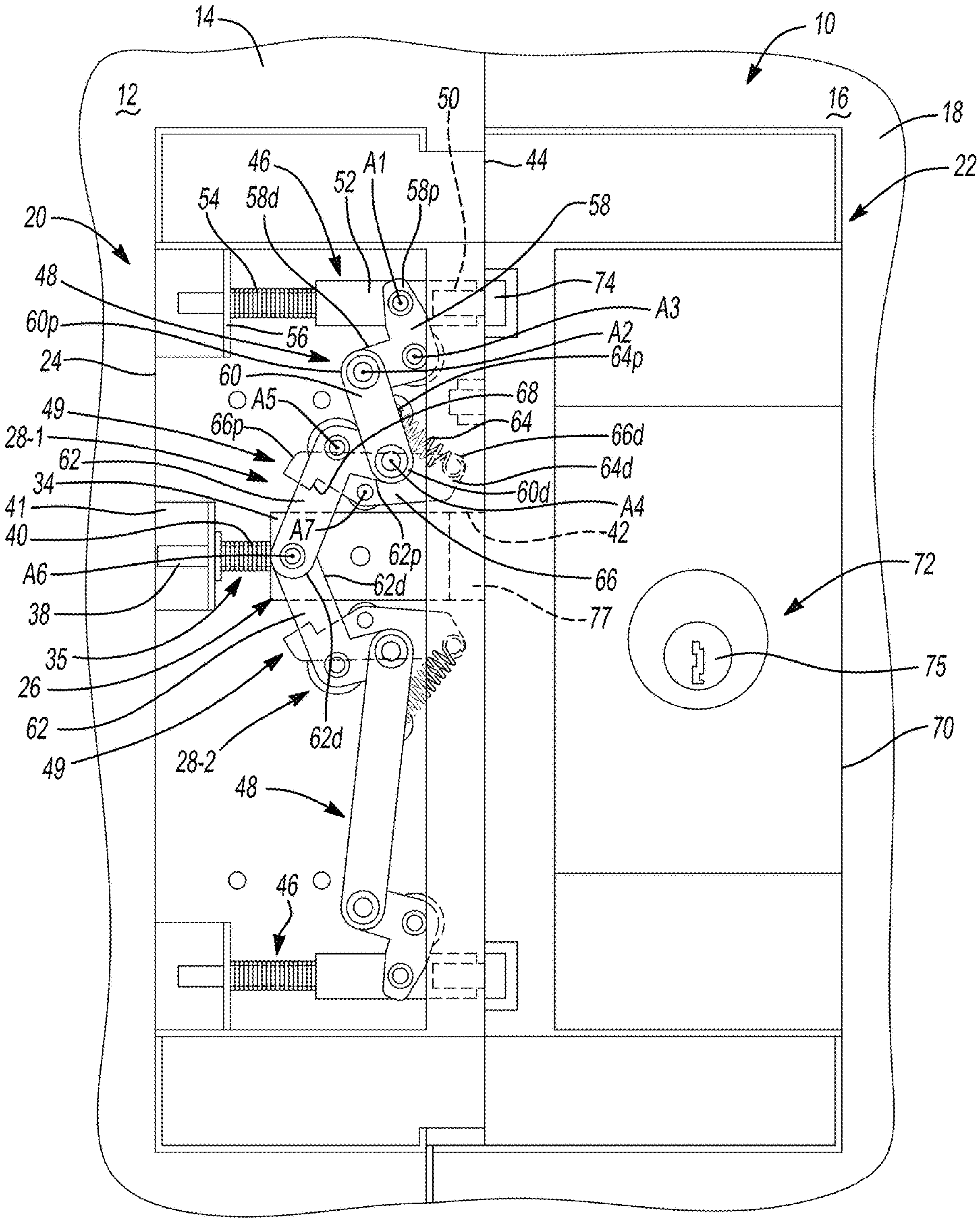
Fig-1A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A2
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A5
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66p
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66d
28-1
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64d
34
60d
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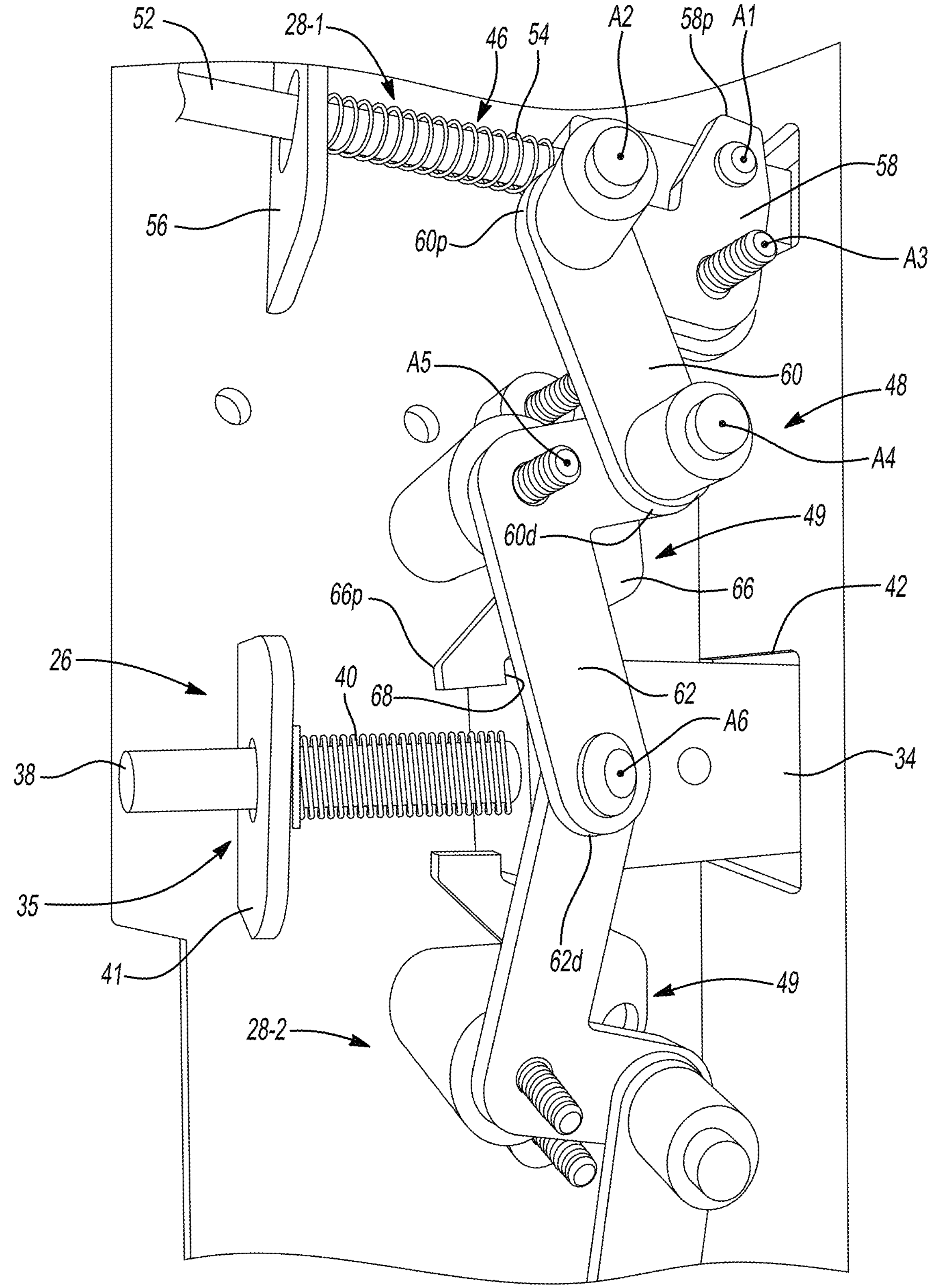
Fig-1B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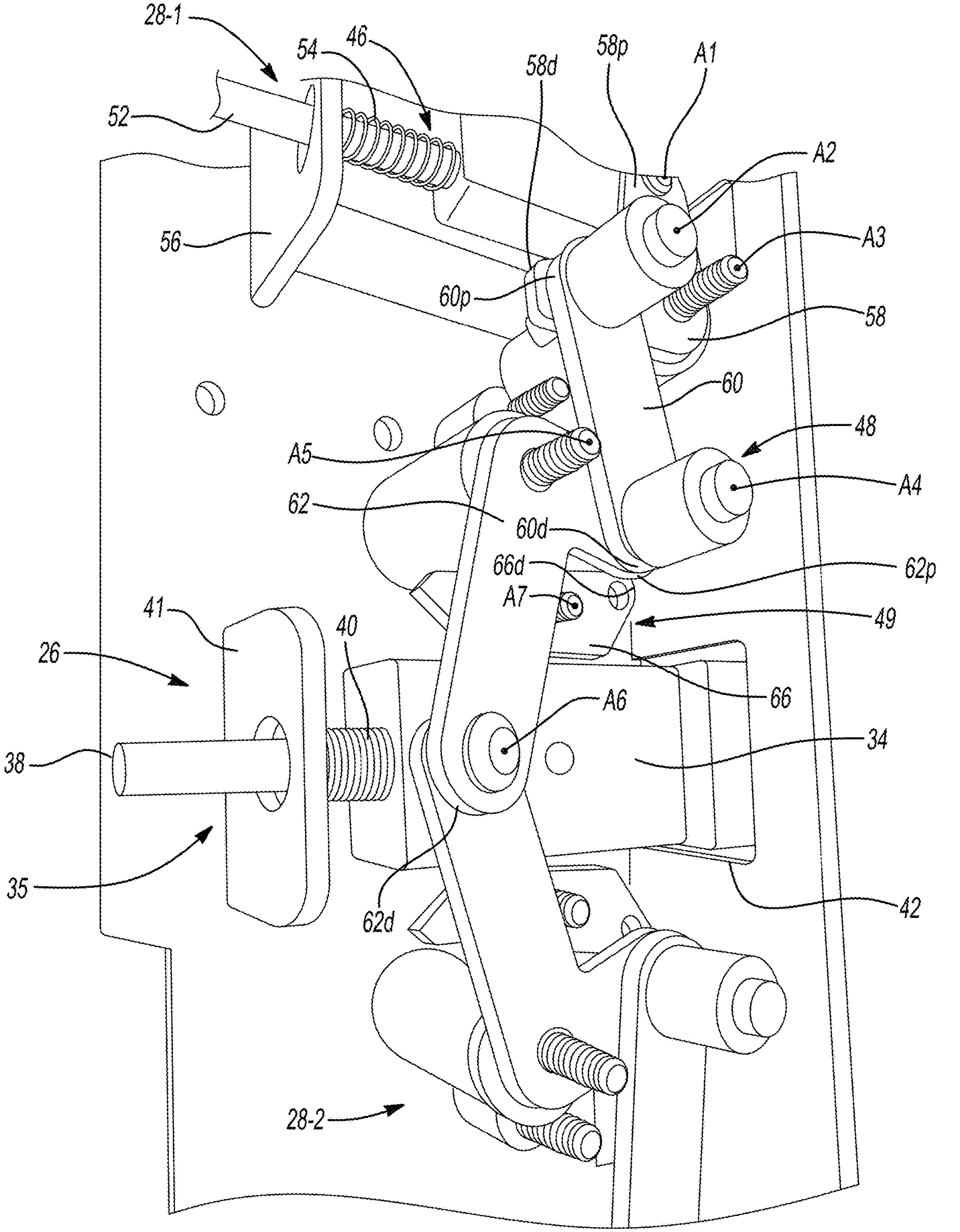
28-2

Fig-2A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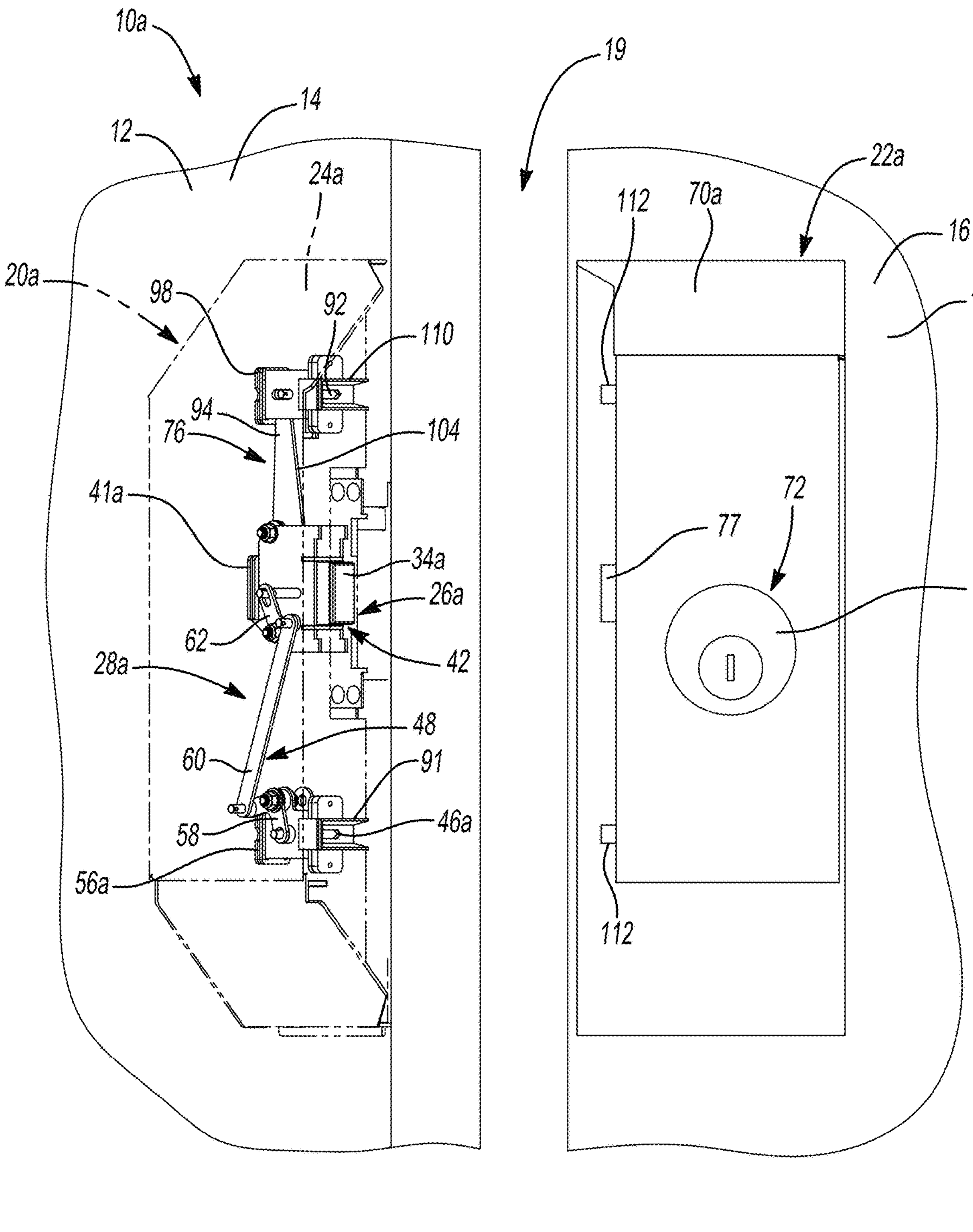
28-2

Fig-2B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56a
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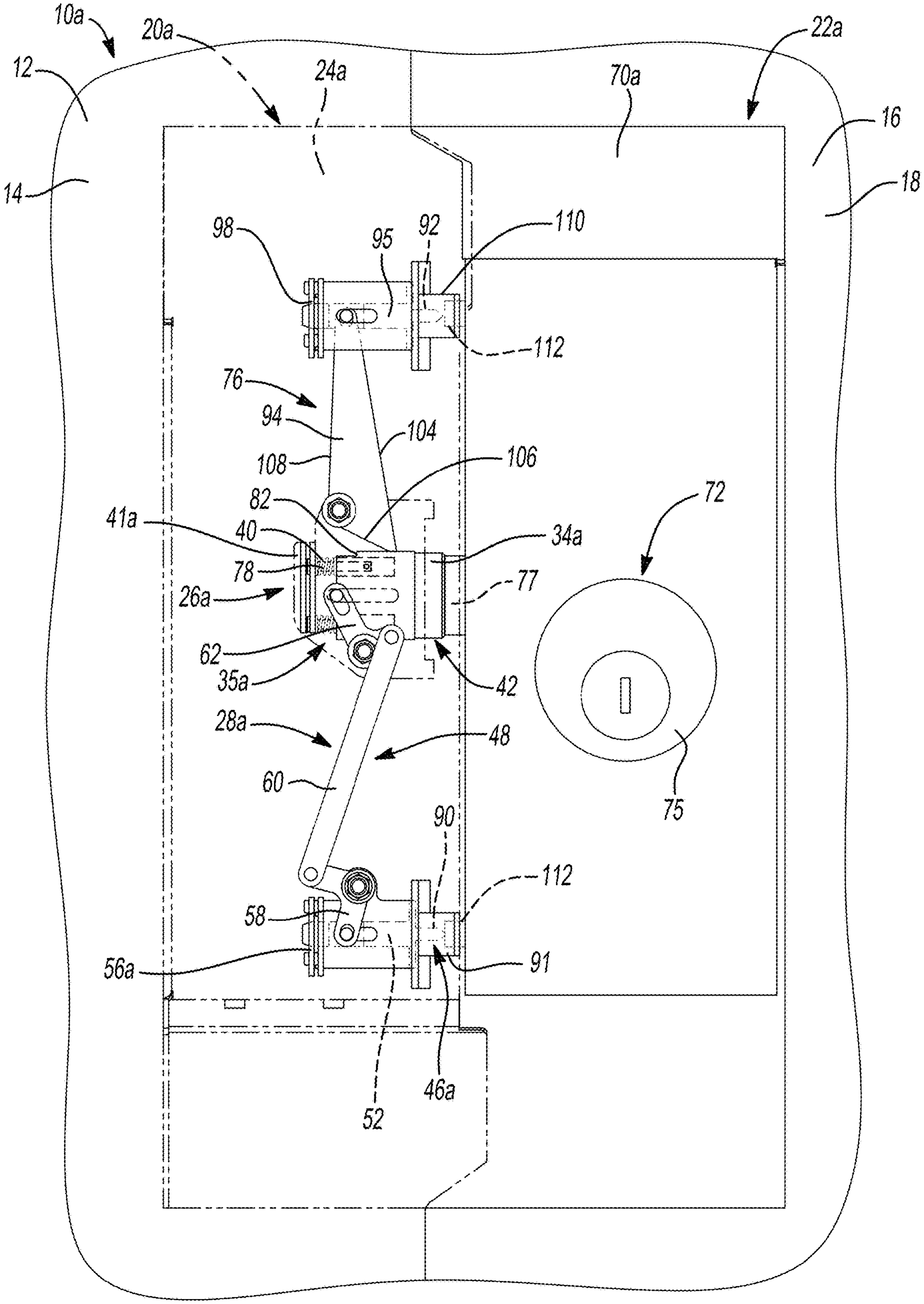
Fig-3A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46a
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Fig-3B

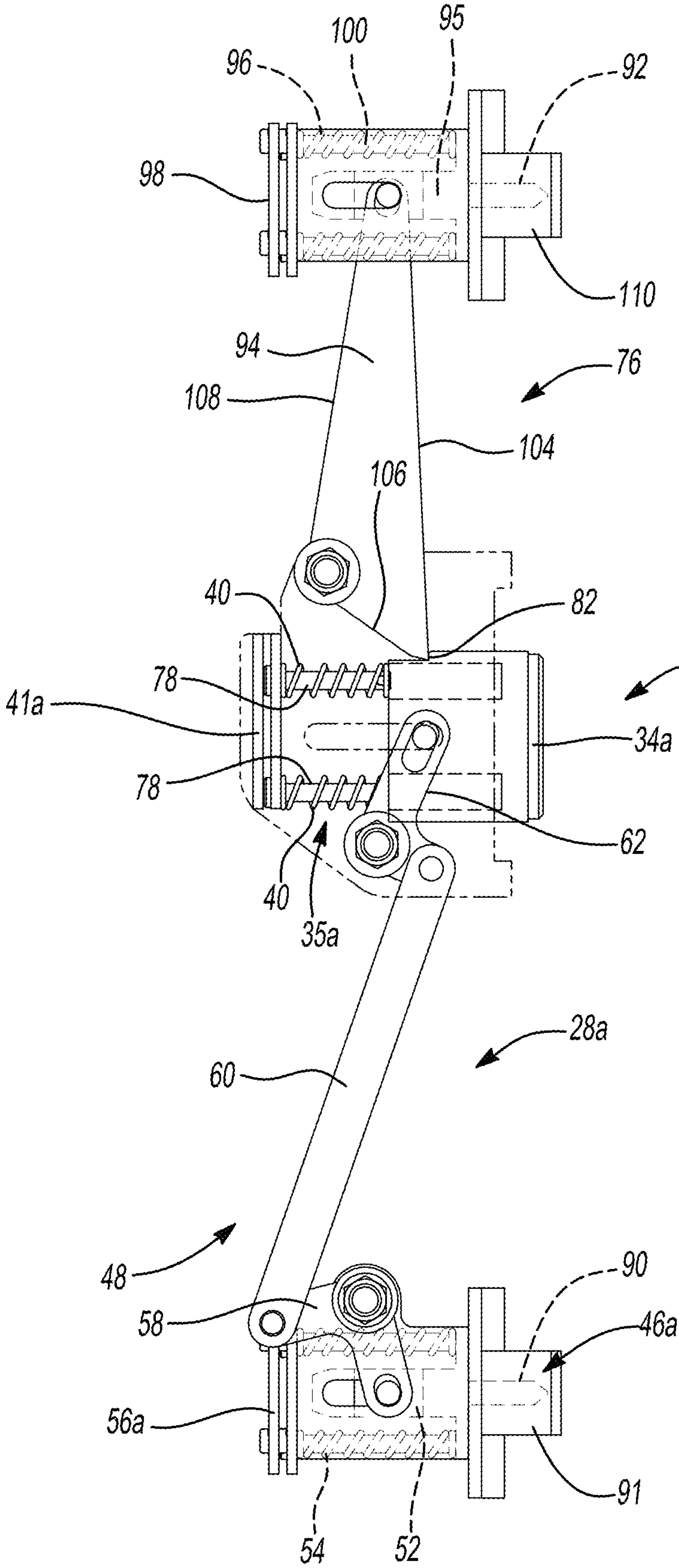
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Fig-4

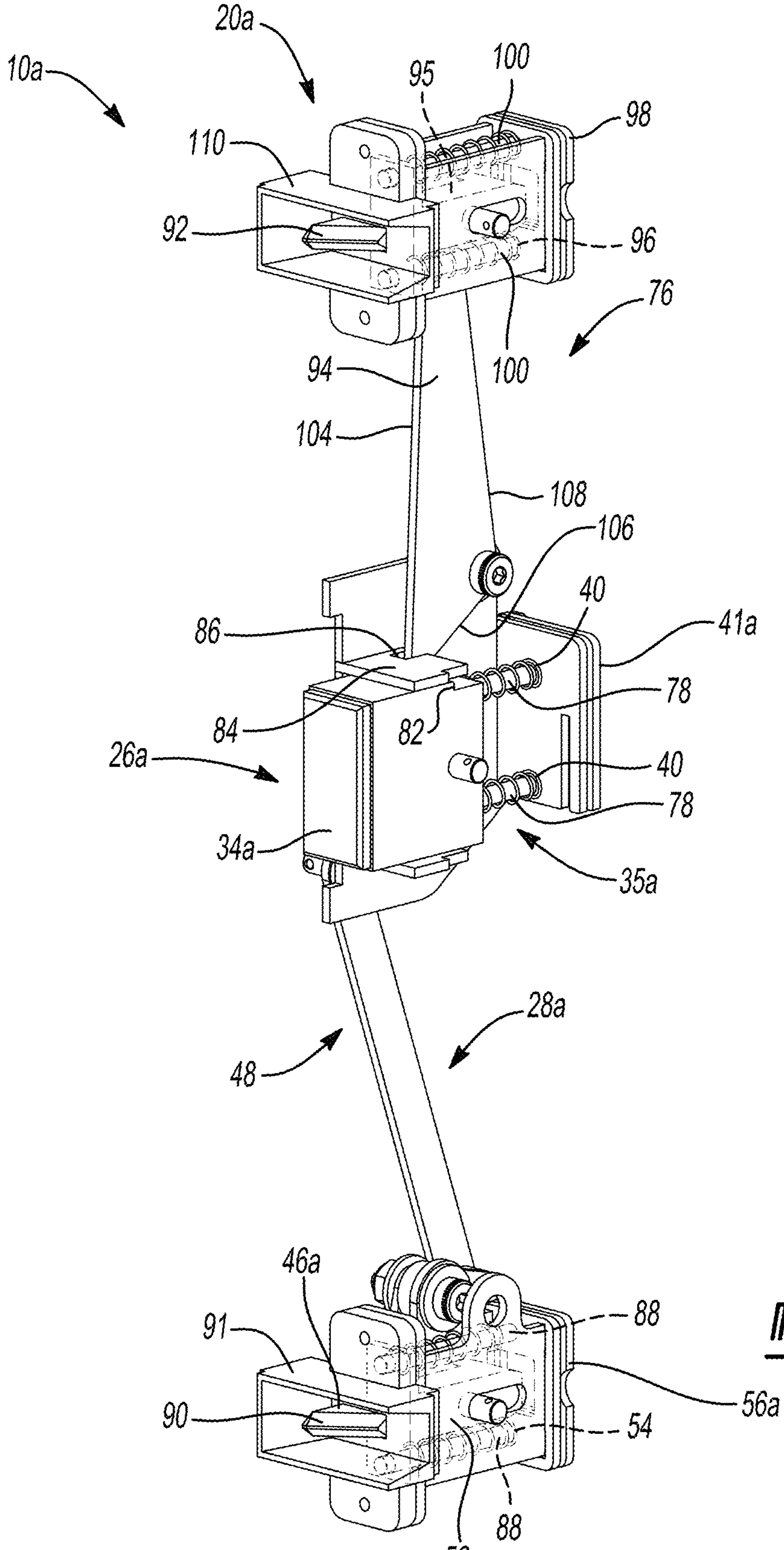
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Fig-5

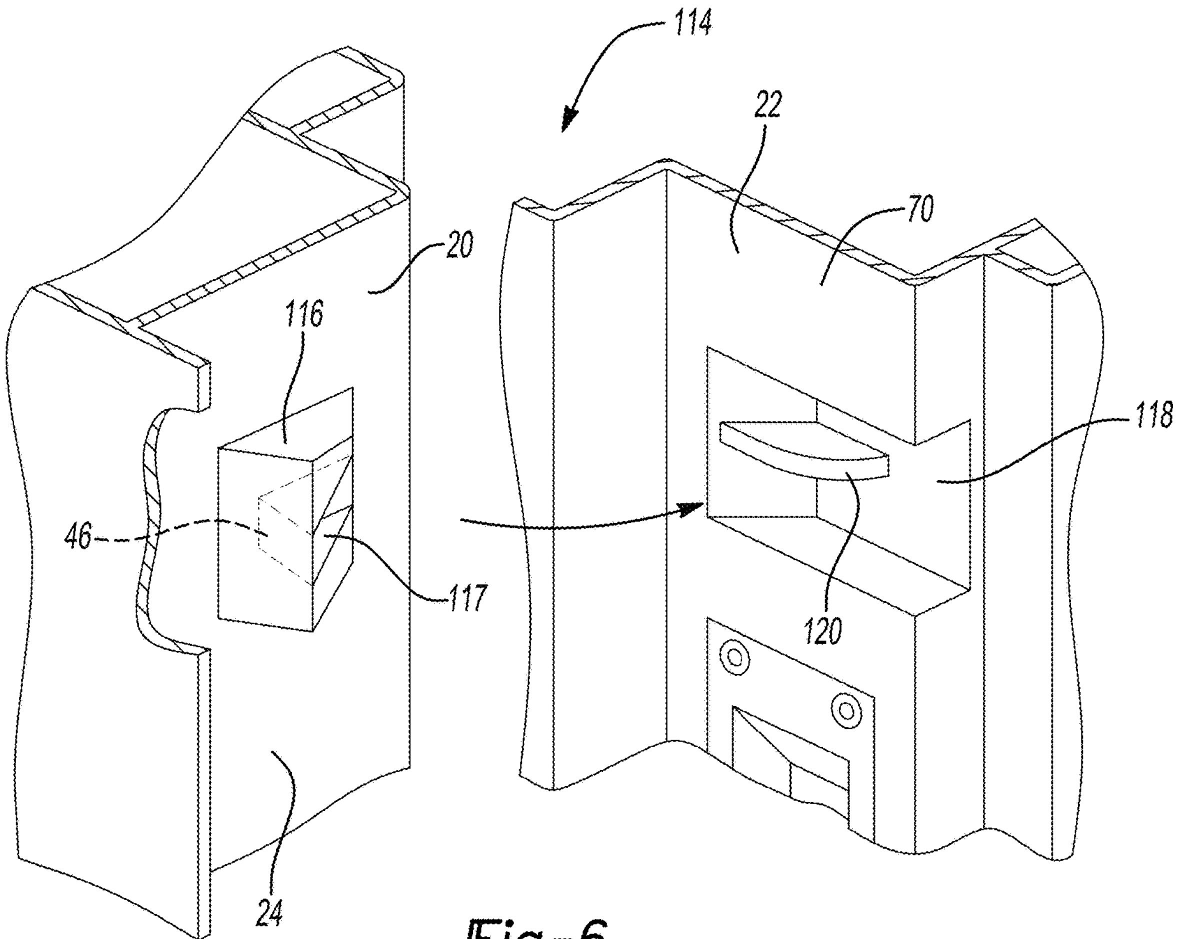


Fig-6

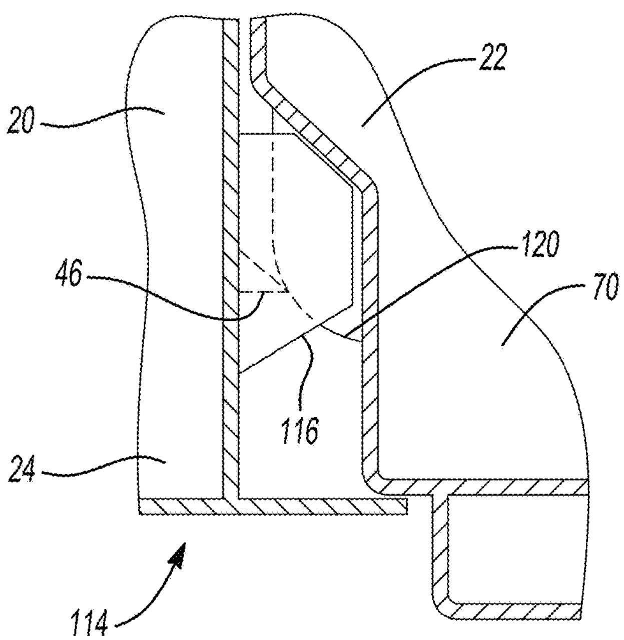
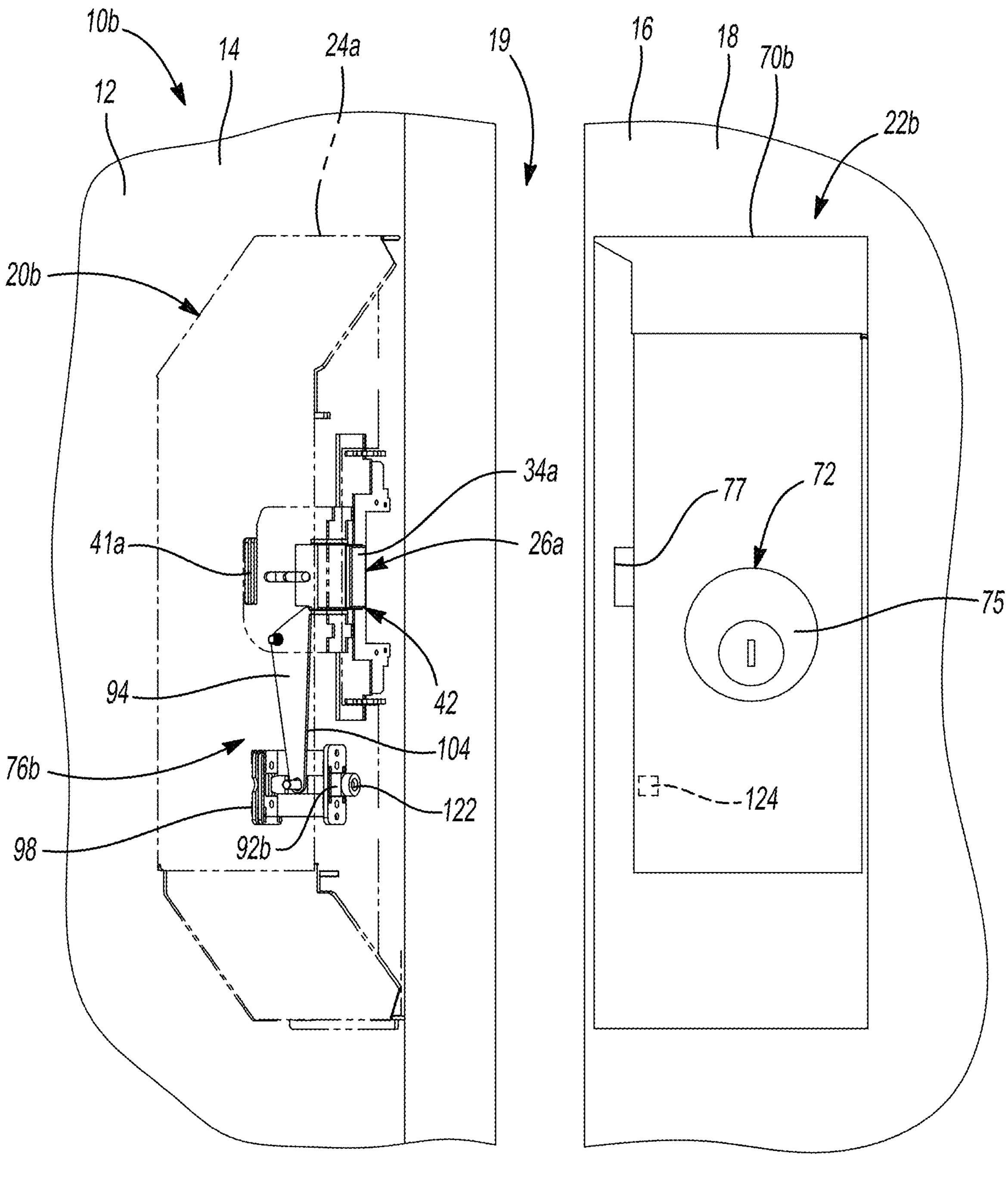


Fig-7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92b

Fig-8A

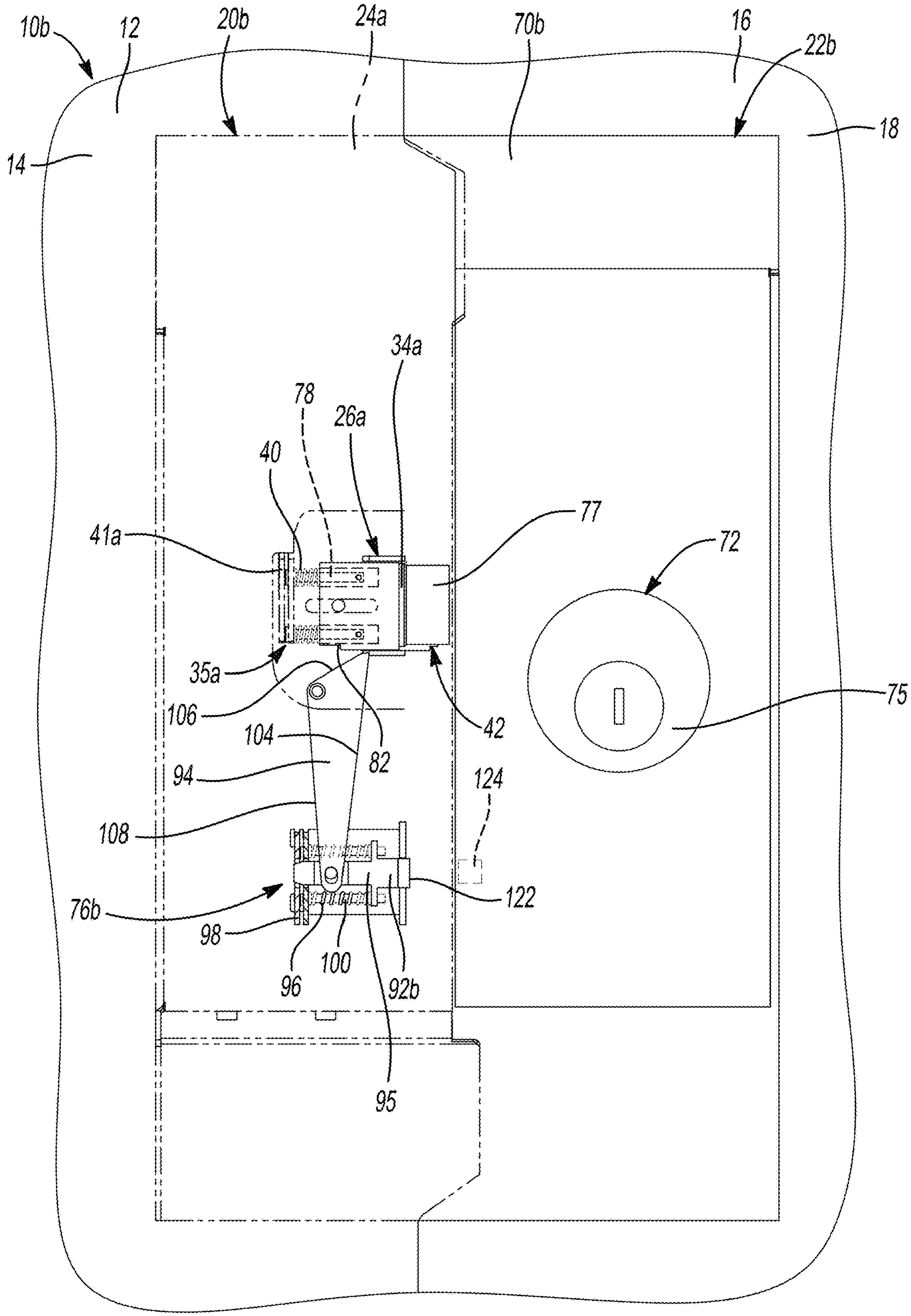
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Fig-8B

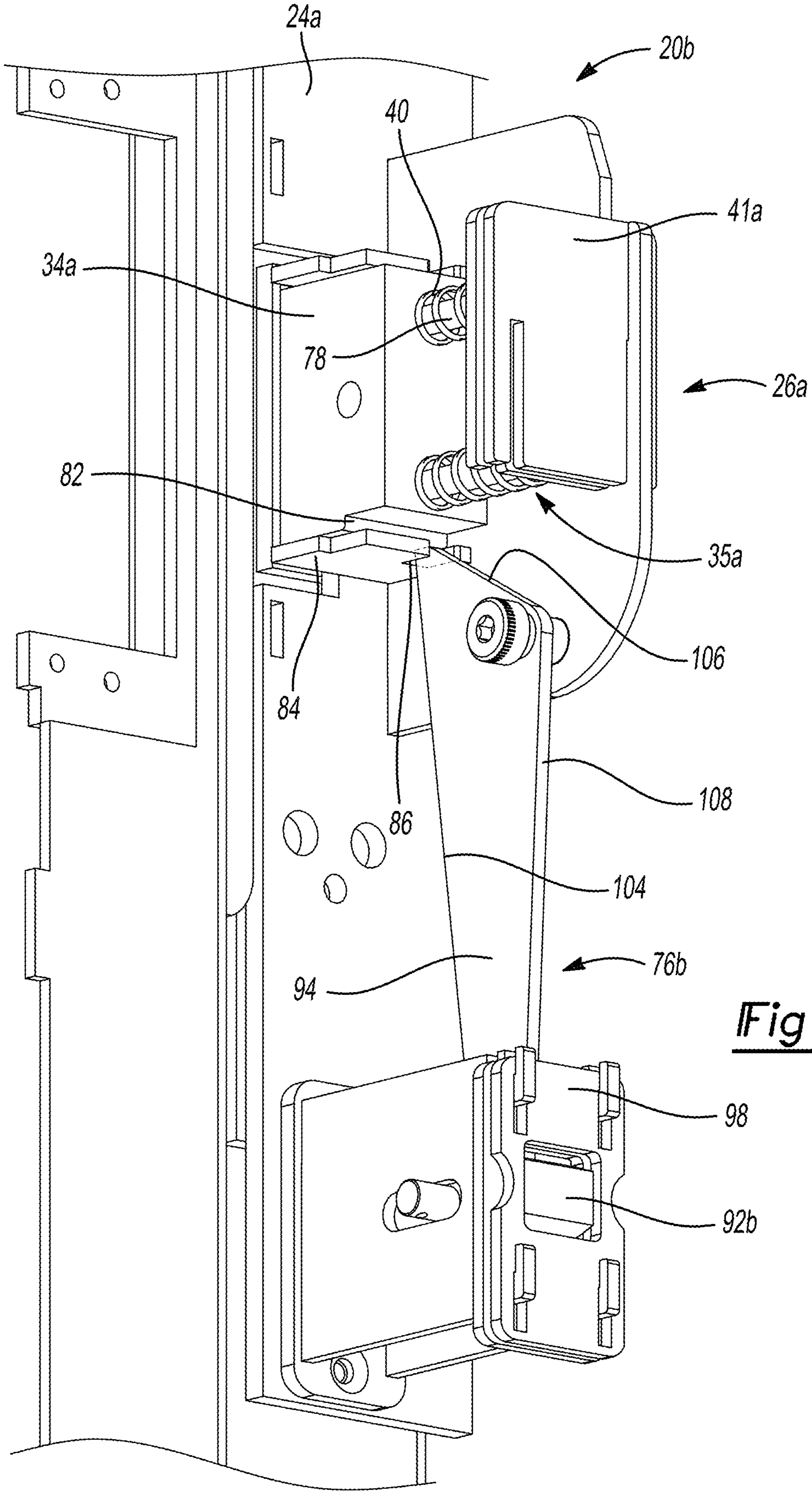

Fig-9

TAMPER-RESISTANT DOOR LOCKING SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to U.S. Provisional Application No. 63/562,721 filed Mar. 8, 2024, the entire disclosure of which is incorporated by reference.

FIELD

[0002] The present disclosure relates to a tamper-resistant locking system for a door and, more particularly, to a locking system including an assembly for plugging an opening formed in a strike plate.

BACKGROUND

[0003] Door locking systems are essential components for securing doors at prisons and various other properties and facilities. These systems are typically attached to the exterior side of a door, allowing authorized individuals to access a room (e.g., a prison cell) on the interior side of the door, and preventing unauthorized individuals from opening the door and exiting the room on the interior side of the door. The security of these door locking systems can be compromised if the housing enclosing the locking system is susceptible to tampering or forced entry. For example, tampering with the door locking system or components thereof can prevent an aperture or opening formed in one of the door or a frame in which the door is received from receiving a lock bolt disposed on the other of the door or the frame, which, in turn, can prevent the door from being secured in a closed position relative to the frame.

[0004] While known systems and methods for securing door locking systems may be acceptable for their intended purposes, there exists a continuous need in the pertinent art for an improved system and method for inhibiting tampering with a door locking system.

[0005] The background description provided here is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

SUMMARY

[0006] One aspect of the disclosure provides a lock system. The lock system includes a first housing, a plug, and a deadlock system. The first housing including a first side defining an opening. The plug is disposed within the first housing and is configured to move within the opening between an open position and a plugged position. The deadlock system is disposed within the first side of the first housing and coupled to the plug. The deadlock system is configured to move between a first position and a second position. The deadlock system (i) inhibits movement of the plug in the first position and (ii) allows movement of the plug in the second position.

[0007] Another aspect of the disclosure provides a lock system. The lock system includes a first housing, a plug, and a first actuation system. The first housing includes a first side defining an opening. The plug is disposed within the first housing and configured to move within the opening between an open position and a plugged position. The first actuation system is disposed within the first side of the first housing and configured to move between a first position and a second position. The first actuation system (i) moves the plug from the plugged position to the open position when the first actuation system moves from the first position to the second position, and (ii) secures the plug in the plugged position when the first actuation system moves from the second position to the first position.

[0008] Further areas of applicability of the present disclosure will become apparent from the detailed description, the claims, and the drawings. The detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The present disclosure will become more fully understood from the detailed description and the accompanying drawings.

[0010] FIG. **1**A is a perspective view of a door locking system in a first orientation according to the principles of the present disclosure.

[0011] FIG. **1**B is a perspective view of the door locking system of FIG. **1**A in a second orientation according to the principles of the present disclosure.

[0012] FIG. **2**A is a second perspective view of the door locking system of FIG. **1**A in the first orientation according to the principles of the present disclosure.

[0013] FIG. **2**B is a perspective view of the door locking system of FIG. **1**A in the second orientation according to the principles of the present disclosure.

[0014] FIG. **3**A is a perspective view of another door locking system in a first orientation according to the principles of the present disclosure.

[0015] FIG. **3**B is a perspective view of the door locking system of FIG. **3**A in a second orientation according to the principles of the present disclosure.

[0016] FIG. **4** is a front view of a first assembly of the door locking system of FIG. **3**A according to the principles of the present disclosure.

[0017] FIG. **5** is a perspective view of the first assembly of FIG. **4** according to the principles of the present disclosure.

[0018] FIG. **6** is a perspective view of a portion of a door locking system in a first orientation according to the principles of the present disclosure.

[0019] FIG. **7** is a top view of the portion of the door locking system of FIG. **6** in a second orientation according to the principles of the present disclosure.

[0020] FIG. **8**A is a perspective view of another door locking system in a first orientation according to the principles of the present disclosure.

[0021] FIG. **8**B is a perspective sectional view of the door locking system of FIG. **8**A in a second orientation according to the principles of the present disclosure.

[0022] FIG. **9** is a perspective view of a first assembly of the door locking system of FIG. **8**A according to the principles of the present disclosure.

[0023] In the drawings, reference numbers may be reused to identify similar and/or identical elements.

DETAILED DESCRIPTION

[0024] Referring to FIGS. **1**A and **1**B, a lock system **10** is illustrated. As will be explained in more detail below, the lock system **10** may be mounted on an exterior surface **12** of a door **14** and an exterior surface **16** of a wall **18** to secure the door **14** in a closed orientation (FIG. **1**B) relative to the wall **18**. The door **14** may pivot (e.g., clockwise) relative to the wall **18** from an open orientation (FIG. **1**A) to the closed orientation (FIG. **1**B) to inhibit passage through an opening **19** defined by the door **14** and the wall **18**. Conversely, the door **14** may pivot relative to the wall **18** from the closed orientation (FIG. **1**B) to the open orientation (FIG. **1**A) to allow passage through the opening **19**.

[0025] The lock system **10** may include a first housing assembly **20** and a second housing assembly **22**. While the first housing assembly **20** is generally shown and described herein as being mounted on the door **14**, and the second housing assembly **22** is generally shown and described herein as being mounted on the wall **18**, it will be appreciated that the first housing assembly **20** may be mounted on the wall **18**, and the second housing assembly **22** may be mounted on the door **14** within the scope of the present disclosure. In some implementations, the first housing assembly **20** may be the door **14** or the wall **18** or be mounted within the door **14** or the wall **18**, and the second housing assembly **22** may be the other of the door **14** or the wall **18** or be mounted within the other of the door **14** or the wall **18**. In this regard, the door **14** may be the first housing assembly **20** or the second housing assembly **22**, and the wall **18** may be the other of the first housing assembly **20** or the second housing assembly **22**.

[0026] The first housing assembly **20** may include a first housing **24**, a plug subassembly **26**, and one or more actuation systems **28**. For example, the first housing assembly **20** may include a first actuation system **28-1**, and a second actuation system **28-2**. The plug subassembly **26**, the first actuation system **28-1**, and the second actuation system **28-2** may be disposed within the first housing **24**. As will be explained in more detail below, the first and/or second actuation systems **28-1**, **28-2** may be configured to inhibit movement of the plug subassembly **26** between a plugged position (FIG. **1**A) and an open position (FIG. **1**B). For example, in a method of operating the lock system **10**, movement of the first actuation system **28-1** and/or the second actuation system **28-2** from a first orientation (FIG. **1**B, **2**B) to a second orientation (FIG. **1**A, **2**A) may move the plug subassembly from the open position (FIG. **1**B) to the plugged position (FIG. **1**A) and inhibit movement of the plug subassembly **26** from the plugged position (FIG. **1**A) to the open position (FIG. **1**B), while movement of the first actuation system **28-1** and/or the second actuation system **28-2** from the second orientation (FIG. **1**A, **2**A) to the first orientation (FIG. **1**B, **2**B) may move (e.g., automatically) the plug subassembly **26** from the plugged position (FIG. **1**A) to the open position (FIG. **1**B). As will be explained in more detail below, engagement of the first actuation system **28-1** and/or the second actuation system **28-2** with the second housing assembly **22** may actuate the first actuation system **28-1** and/or the second actuation system **28-2** from the second orientation (FIG. **1**A) to the first orientation (FIG. **1**B) and move the plug subassembly **26** from the plugged position (FIG. **1**A) to the open position (FIG. **1**B). Conversely, disengagement of the first actuation system **28-1** and/or the second actuation system **28-2** from the second housing assembly **22** may move (e.g., automatically) the plug subassembly **26** to the plugged position (FIG. **1**A). When the lock system **10** is in the open orientation (FIG. **1**A), the plug subassembly **26** may be in the plugged position (FIG. **1**A). When the lock system **10** is in the closed orientation (FIG. **1**B), the plug subassembly **26** may be in the open position (FIG. **1**B).

[0027] The plug subassembly **26** may be coupled to the first housing **24** and may include a plug **34** and a plug actuation system **35**. In some implementations, the plug actuation system **35** includes a post **38** and a biasing member **40**. As illustrated in FIG. **1**B, the post **38** may be coupled to the plug **34** and translatably-coupled to the first housing **24**. For example, the first housing **24** may include a bracket **41** to which the post **38** is translatably-coupled. In particular, the bracket **41** may define an aperture through which the post **38** is disposed. The biasing member **40** (e.g., a compression spring) may include a first end engaging the plug **34** and a second end, opposite the first end, engaging the bracket **41**.

[0028] As will be explained in more detail below, during operation of the lock system **10**, the plug actuation system **35** may urge movement of the plug **34** from an open position (FIG. **1**B) to a plugged position (FIG. **1**A). For example, upon moving the door **14** from the closed position (FIG. **1**B) to the open position (FIG. **1**A), the biasing member **40** may apply a force on the plug **34** and the bracket **41** to urge the plug **34** and the post **38** to translate relative to the first housing **24** (e.g., toward the second housing assembly **22**). In the plugged position (FIG. **1**A), the plug **34** may extend outside of the first housing **24**. In the open position (FIG. **1**B), the plug **34** may be retracted within the first housing **24**. When the door **14** is in the open orientation (FIG. **1**A), the plug **34** may be in the plugged position (FIG. **1**A), and when the door **14** is in the closed orientation (FIG. **1**B), the plug **34** may be in the open position (FIG. **1**B).

[0029] The first housing **24** may define an opening **42** disposed on a first side **44** of the first housing **24**. When the door **14** is in the closed orientation (FIG. **1**B), the first side **44** of the first housing **24** may face the second housing assembly **22**. The opening **42** may receive the plug **34**. In this regard, the opening **42** may be substantially the same shape and/or size as at least a portion of the plug **34**. In some configurations, the opening **42** may be a rectangle. The opening **42** may be slightly larger than the plug **34** to allow the plug **34** to move (e.g., translate) freely through the opening **42**. In some configurations, the opening **42** may be sized such that there is a zero or near zero clearance between the plug **34** and the opening **42**.

[0030] With reference to FIGS. **1**A-**2**B, the first actuation system **28-1** may include an actuator **46**, a linkage assembly **48**, and a deadlock assembly **49**. The actuator **46** may include a wheel **50**, post **52**, and a biasing member **54**. As illustrated in FIG. **1**B, the wheel **50** may be rotatably-coupled to the post **52**, and the post **52** may be translatably-coupled to the first housing **24**. For example, the first housing **24** may include a bracket **56** to which the post **52** is translatably-coupled. For example, the bracket **56** may define an aperture through which the post **52** is disposed. The biasing member **54** (e.g., a compression spring) may include a first end engaging the post **52** and a second end, opposite the first end, engaging the bracket **56**.

[0031] When the lock system **10** is in the open orientation (FIG. **1**A), the actuator **46** (e.g., the wheel **50**) may extend outside the first housing **24**. For example, the wheel **50** may extend beyond the first side **44** of the first housing **24**. When the lock system **10** is in the closed orientation (FIG. **1**B) the actuator **46** may be disposed inside the first housing **24**. In this regard, as will be explained in more detail below, during operation of the lock system **10**, upon moving the door **14** from the closed position (FIG. **1**B) to the open position (FIG. **1**A), the biasing member **54** may apply a force on the post **52** and the bracket **56** to move the wheel **50** and the post **52** relative to the first housing **24** (e.g., toward the second housing assembly **22**).

[0032] With reference to FIGS. **2**A and **2**B, the linkage assembly **48** may include a first link **58**, a second link **60**, and a third link **62**. The first link **58**, the second link **60**, and the third link **62** may each include a proximal end and a distal end. The first link **58** (e.g., the proximal end **58***p*) may be rotatably coupled to the actuator **46** (e.g., the post **52**) for rotation about a first axis A1. The first link **58** (e.g., the distal end **58***d*) may be rotatably coupled to the second link **60** (e.g., the proximal end **60***p*) for rotation about a second axis A2. The first link **58** may be further rotatably coupled to the first housing **24** for rotation about a third axis A3.

[0033] The second link **60** (e.g., the distal end **60***d*) may be rotatably coupled to the third link **62** (e.g., the proximal end **62***p*) for rotation about a fourth axis A4. The third link **62** may be rotatably coupled to the first housing **24** for rotation about a fifth axis A5. The third link **62** may be further rotatably coupled to the plug **34** or a portion (e.g., post **38**) of the plug actuation system **35** for rotation about a sixth axis A6.

[0034] The first link **58**, the second link **60**, and the third link **62** may each move (e.g., rotate) about a rotation feature (e.g., a hub) that defines the axes A1, A2, A3, A4, A5, A6, respectively. As the actuator **46** is forced into the first housing **24**, the first link **58** may rotate in a counterclockwise direction about the third axis A3, and the third link **62** may rotate in a clockwise direction about the fifth axis A5 and force the plug **34** to move (e.g., translate) within the opening **42** from the plugged position (FIG. **1**A) to the open position (FIG. **1**B).

[0035] With reference to FIG. **1**B, the deadlock assembly **49** may include a biasing member **64** (e.g., spring) and a link **66**. The link **66** may include a proximal end **66***p* and a distal end **66***d*. The proximal end **66***p* may include a substantially planar locking surface **68**. The link **66** may be rotatably coupled to the first housing **24** for rotation about an axis A7. For example, the link **66** may move (e.g., rotate) about a rotation feature (e.g., a hub) that defines the axis A7.

[0036] The biasing member **64** may include a proximal end **64***p* coupled to one of the second link **60** or the first housing **24**, and a distal end **64***d* coupled to the link **66** (e.g., the distal end **66***d*). In some implementations, the biasing member **64** may be coupled to the second link **60** between the proximal and distal ends **60***p*, **60***d* thereof. As will be explained in more detail below, the biasing member **64** may apply a force on the link **66** urging counterclockwise rotation of the link **66** about the seventh axis A7. When the door **14** is in the open orientation (FIG. **2**A), and the plug subassembly **26** is in the plugged position (FIG. **1**A), the surface **68** may contact the plug **34** to further inhibit translation of the plug **34** relative to the first housing **24** when a force is applied on the plug **34**.

[0037] The second actuation system **28-2** may include components (e.g., an actuator **46**, a linkage assembly **48**, and a deadlock assembly **49**) having a similar structure and function as those described above relative to the first actuation system **28-1**. In view of the similarity in structure and function of the first actuation system **28-1** and the second actuation system **28-2**, the second actuation system **28-2** will not be separately described herein, and the description of the first actuation system **28-1** will be understood to apply equally to the second actuation system **28-2** except as otherwise noted. In some implementations, the third link **62** (e.g., the distal end **62***d*) of the first actuation system **28-1** may be rotatably coupled to the third link **62** (e.g., the distal end **62***d*) of the second actuation system **28-2** for rotation about the sixth axis A6 such that the third link **62** of the first actuation system **28-1** can rotate about the fifth axis A5 of the first actuation system **28-1** upon rotation of the third link **62** of the second actuation system **28-1** about the fifth axis A5 of the second actuation system **28-1**. In this way, the actuator **46** of the first actuation system **28-1** or the actuator **46** of the second actuation system **28-2** may move from the first position (FIG. **1**A) to the second position (FIG. **1**B) upon simultaneous movement of the actuator **46** of the first actuation system **28-1** and the actuator **46** of the second actuation system **28-2** from the first position (FIG. **1**A) to the second position (FIG. **1**B).

[0038] The second housing assembly **22** may include a second housing **70** and a locking system **72** coupled to the second housing **70**. The second housing **70** may include one or more ramps **74**. The ramps **74** may be configured to receive the actuator **46** (e.g., the wheel **50**) when the door **14** is in the closed orientation (FIG. **1**B). The locking system **72** may include a lock mechanism **75** and a latch **77**. The latch **77** may secure the position of the first housing assembly **20** relative to the second housing assembly **22** when the door **14** is in the closed orientation. For example, when the door **14** is in the closed orientation (FIG. **1**B), the latch **77** may be disposed within the opening **42** of the first housing **24**.

[0039] Referring now to FIGS. **3**A-**5**, a lock system **10***a* is illustrated. In view of the substantial similarity in structure and function of the components associated with the lock system **10***a* with respect to the lock system **10**, like reference numerals are used hereinafter, and in the drawings, to identify like components while like reference numerals containing letter extensions (e.g., "a") are used to identify those components that have been modified. Any references herein to the lock system **10**, or any component thereof, will be understood to apply equally to the lock system **10***a*, or the corresponding component thereof, unless otherwise noted.

[0040] The lock system **10***a* may include a first housing assembly **20***a* and a second housing assembly **22***a*. The first housing assembly **20***a* may include a first housing **24***a*, a plug subassembly **26***a*, an actuation system **28***a*, and a deadlock system **76**. The plug subassembly **26***a*, the actuation system **28***a*, and the deadlock system **76** may be disposed within the first housing **24***a*. As will be explained in more detail below, the actuation system **28***a* may allow and/or inhibit movement of the plug subassembly **26***a* between a plugged position (FIG. **3**A) and an open position (FIG. **3**B). For example, in a method of operating the lock system **10***a*, movement of the actuation system **28***a* from a first orientation (FIG. **3**B) to a second orientation (FIG. **3**A) may move the plug subassembly **26***a* from the open position (FIG. **3**B) to the plugged position (FIG. **3**A) and inhibit movement of the plug subassembly **26***a* from the plugged position (FIG. **3**A) to the open position (FIG. **3**B), while movement of the actuation system **28***a* from the second orientation (FIG. **3**A) to the first orientation (FIG. **3**B) may move (e.g., automatically) the plug subassembly **26***a* from the plugged position (FIG. **3**A) to the open position (FIG. **3**B).

[0041] As will be explained in more detail below, engagement of the actuation system **28***a* with the second housing assembly **22***a* may actuate the actuation system **28***a* from the second orientation (FIG. **3**A) to the first orientation (FIG. **3**B) and move the plug subassembly **26***a* from the plugged position (FIG. **3**A) to the open position (FIG. **3**B). Conversely, disengagement of the actuation system **28***a* from the second housing assembly **22***a* may move (e.g., automatically) the plug subassembly **26***a* to the plugged position (FIG. **3**A). When the lock system **10***a* is in an open orientation (FIG. **3**A), the plug subassembly **26***a* may be in the plugged position (FIG. **3**A). When the lock system **10***a* is in a closed orientation (FIG. **3**B), the plug subassembly **26***a* may be in the open position (FIG. **3**B).

[0042] The plug subassembly **26***a* may be coupled to the first housing **24***a* and may include a plug **34***a* and a plug actuation system **35***a*. In some implementations, the plug actuation system **35***a* includes one or more biasing members **40**, and one or more guide rods **78**. As illustrated in FIG. **3**B, each of the one or more guide rods **78** may be coupled to the first housing **24***a* and translatably-coupled to the plug **34***a*. For example, the first housing **24***a* may include a bracket **41***a* to which each of the one or more guide rods **78** is coupled. The plug **34***a* may define one or more apertures and each of the one or more guide rods **78** may be disposed through a corresponding aperture of the one or more apertures. A biasing member **40** of the one or more biasing members **40** may be disposed around each of the one or more guide rods **78**. As will be explained in more detail below, as the plug **34***a* moves from the plugged position (FIG. **3**A) to the open position (FIG. **3**B), or vice versa, the plug **34***a* may translate (e.g., slide) along the one or more guide rods **78**.

[0043] In some implementations, the plug **34***a* includes a notch **82** in one or more sides of the plug **34***a*. The notch **82** may engage and disengage one or more components of the deadlock system **76** to allow and/or inhibit movement of the plug subassembly **26***a* (e.g., the plug **34***a*). In some implementations, the plug subassembly **26***a* includes a retaining clip **84** coupled to the plug **34***a*. The retaining clip **84** may include a slot **86** that receives one or more components of the deadlock system **76**.

[0044] The actuation system **28***a* may include an actuator **46***a* and the linkage assembly **48**. The actuator **46***a* may include the post **52**, one or more guide rods **88** and one or more biasing members **54**. The one or more guide rods **88** may be coupled to the first housing **24***a* and translatably-coupled to the actuator **46***a*. For example, the first housing **24***a* may include a bracket **56***a* to which one or more guide rods **88** are coupled. A biasing member **54** of the biasing members **54** may be disposed around each of the one or more guide rods **88**. In some implementations, the bracket **56***a* defines an aperture through which the post **52** is disposed. In some implementations, the actuator **46***a* defines one or more apertures and each of the one or more guide rods **88** is disposed through a corresponding aperture of the one or more apertures. In some implementations, the actuator **46***a* includes a chamfered (e.g., sharpened) end **90**. When the lock system **10***a* is in the open orientation (FIG. **3**A), the chamfered end **90** may deter a user from engaging, or otherwise applying a force on, the actuator **46***a*.

[0045] When the lock system **10***a* is in the open orientation (FIG. **3**A), the actuator **46***a* (e.g., the chamfered end **90**) may extend outside the first housing **24***a*. For example, the chamfered end **90** may extend beyond the first side **44** of the first housing **24***a*. When the lock system **10***a* is in the closed orientation (FIG. **3**B) the actuator **46***a* may be disposed inside the first housing **24**. In this regard, as will be explained in more detail below, during operation of the lock system **10***a*, upon moving the door **14** from the closed position (FIG. **3**B) to the open position (FIG. **3**A), the biasing member(s) **54** may apply a force on the actuator **46***a* and the bracket **56***a* to move the actuator **46***a* relative to the first housing **24***a* (e.g., toward the second housing assembly **22***a*).

[0046] In some implementations, the actuation system **28***a* includes a covering **91** that at least partially covers the actuator **46***a*. For example, when the door **14** is in the open position (FIG. **3**A), the covering **91** may prevent an individual from moving or tampering with the actuator **46***a*. When the door **14** is in the closed position (FIG. **3**B), the covering **91** may receive one or more components of the second housing assembly **22***a*.

[0047] The deadlock system **76** may include an actuator **92** and a locking lever **94**. As will be described in more detail below, the deadlock system **76** may be configured to allow and/or inhibit movement of the plug subassembly **26***a* between the plugged position (FIG. **3**A) and the open position (FIG. **3**B). For example, in a first orientation (FIG. **3**A), the deadlock system **76** may engage the plug subassembly **26***a* and prevent the plug subassembly **26***a* from moving from the plugged position (FIG. **3**A) to the open position (FIG. **3**B). In a method of operating the lock system **10***a*, movement of the deadlock system **76** from the first orientation (FIG. **3**A) to a second orientation (FIG. **3**B) may disengage the deadlock system **76** from the plug subassembly **26***a* and allow the plug subassembly **26***a* to move from the plugged position (FIG. **3**A) to the open position (FIG. **3**B).

[0048] The actuator **92** may include a post **95**, one or more guide rods **100** and one or more biasing members **96**. The one or more guide rods **100** may be coupled to the first housing **24***a* and translatably-coupled to the actuator **92**. For example, the first housing **24***a* may include a bracket **98** to which one or more guide rods **100** are coupled. A biasing member **96** of the biasing members **96** may be disposed around each of the one or more guide rods **100**. In some implementations, the bracket **98** defines an aperture through which the post **95** is disposed. In some implementations, the actuator **92** defines one or more apertures and each of the one or more guide rods **100** is disposed through a corresponding aperture of the one or more apertures. In some implementations, the actuator **92** includes a chamfered (e.g., sharpened) end **102**. When the lock system **10***a* is in the open orientation (FIG. **3**A), the chamfered end **102** may deter a user from engaging, or otherwise applying a force on, the actuator **92**. When the lock system **10***a* is in the open orientation (FIG. **3**A), the actuator **92** (e.g., the chamfered end **102**) may extend outside the first housing **24**. For example, the chamfered end **102** may extend beyond the first side **44** of the first housing **24**.

[0049] The locking lever **94** may be coupled (e.g., rotatably-coupled) to the actuator **92** and may have a substantially triangular shape. In this regard, the locking lever **94** may include a first edge **104**, a second edge **106** transverse to the first edge **104**, and a third edge **108** transverse to the first and second edges **104**, **106**. The first, second, and third edges **104**, **106**, **108** may define the substantially triangular shape of the locking lever **94**. When the deadlock system **76** is in the first orientation (FIG. **3**A), the locking lever **94** may be disposed within the slot **86** of the retaining clip **84** and the first edge **104** may engage the notch **82** of the plug **34***a*. When the deadlock system **76** is in the second orientation (FIG. **3**B), the locking lever **94** may be disengaged from and disposed above the retaining clip **84**.

[0050] As will be explained in more detail below, during operation of the lock system **10***a*, upon movement of the deadlock system **76** from the first orientation (FIG. **3**A) to the second orientation (FIG. **3**B), the locking lever **94** may begin to rotate (e.g., counterclockwise) and apply a force on the retaining clip **84** and/or the notch **82**, which in turn may cause the plug **34***a* to extend further through the opening **42**. The movement of the plug **34***a* may allow the locking lever **94** to disengage from the plug **34***a* (e.g., the notch **82** and/or the retaining clip **84**) and continue to rotate (e.g., counterclockwise) until the locking lever **94** is disposed above the plug **34***a* (FIG. **3**B). Once the locking lever **94** is in the second orientation (FIG. **3**B), the plug **34***a* is free to retract within the first housing **24** (e.g., by movement of the actuation system **28***a*).

[0051] In some implementations, the deadlock system **76** includes a covering **110** that at least partially covers the actuator **92**. For example, when the door **14** is in the open position (FIG. **3**A), the covering **110** may prevent an individual from moving or tampering with the actuator **92**. When the door **14** is in the closed position (FIG. **3**B), the covering **110** may receive one or more components of the second housing assembly **22***a*.

[0052] The second housing assembly **22***a* may include a second housing **70***a* and the locking system **72**. The locking system **72** may be coupled to the second housing **70***a*. The second housing **70***a* may include one or more protrusions **112**. The one or more protrusions **112** may extend from the second housing **70***a* towards the first housing **24***a*. The one or more protrusions **112** may engage the actuator **46***a* and/or the actuator **92** when the door **14** is in the closed position (FIG. **3**B). The one or more protrusions **112** may be disposed within the covering **91** and/or the covering **110** when the door **14** is in the closed position (FIG. **3**B). In some implementations, each of the one or more protrusions **112** may include a tapered end **113**.

[0053] Referring now to FIGS. **6** and **7**, an activation system **114** is illustrated. The activation system **114** may be configured to work with the lock systems **10**, **10***a*. For example, the activation system **114** may activate an actuation system (e.g., actuation system **28**, **28***a*).

[0054] The activation system **114** may include one or more protrusions **116** extending from a first housing (e.g., the first housing **24**, **24***a*). The one or more protrusions **116** may surround at least a portion of the actuators **46**, **46***a*, **92**. In this regard, each protrusion **116** may define a cavity **117**. When the door **14** is in the open position (FIGS. **1**A, **3**A), each actuator **46**, **46***a*, **92** may disposed in one of the cavities **117**. When the door **14** is in the open position (FIGS. **1**A, **3**A), the one or more protrusions **116** may prevent an individual from moving or tampering with the actuators **46**, **46***a*, **92**. In some implementations, there is zero or near zero clearance between the one or more protrusions **116** and the actuators **46**, **46***a*, **92**.

[0055] The activation system **114** may include one or more recesses **118** formed in a second housing (e.g., the second housing **70**, **70***a*). In some implementations, the one or more recesses **118** may replace the one or more ramps **74** or the one or more protrusions **112**. Each recess of the one or more recesses **118** may include a striker **120** extending from the recess **118**. As illustrated in FIG. **6**, the striker **120** may include a convex outer surface. When the door **14** is in the closed position (FIGS. **1**B, **3**B), each striker **120** (e.g., the convex outer surface) may engage and/or move (e.g., translate, roll, etc.) along one of the actuators **46**, **46***a*, **92**. In this regard, each protrusion **116** may disposed in one of the recesses **118** and each of the strikers **120** may be disposed in one of the cavities **117** when the door **14** is in the closed position (FIGS. **1**B, **3**B). Engagement of the striker **120** with the actuator **46**, **46***a*, **92** may urge movement (e.g., translation) of the actuator **46**, **46***a*, **92** within the first housing **24**, **24***a*.

[0056] Referring now to FIGS. **8**A, **8**B, and **9**, a lock system **10***b* is illustrated. In view of the substantial similarity in structure and function of the components associated with the lock system **10***b* with respect to the lock systems **10**, **10***a*, like reference numerals are used hereinafter, and in the drawings, to identify like components while like reference numerals containing letter extensions (e.g., "b") are used to identify those components that have been modified. Any references herein to the lock system **10** or **10***a*, or any component thereof, will be understood to apply equally to the lock system **10***b*, or the corresponding component thereof, unless otherwise noted.

[0057] The lock system **10***b* may include a first housing assembly **20***b* and a second housing assembly **22***b*. The first housing assembly **20***a* may include the first housing **24***a*, the plug subassembly **26***a*, and a deadlock system **76***b*. The plug subassembly **26***a* and the deadlock system **76***b* may be disposed within the first housing **24***a*. The deadlock system **76***b* may include an actuator **92***b* and the locking lever **94**. As will be described in more detail below, the deadlock system **76***b* may be configured to allow and/or inhibit movement of the plug subassembly **26***a* between the plugged position (FIG. **8**A) and the open position (FIG. **8**B). For example, in a first orientation (FIG. **8**A), the deadlock system **76***b* may engage the plug subassembly **26***a* (e.g., the plug **34** or the notch **82**) and prevent the plug subassembly **26***a* from moving from the plugged position (FIG. **8**A) to the open position (FIG. **8**B).

[0058] The actuator **92***b* may include the post **95**, the one or more guide rods **100**, the one or more biasing members **96**, and a magnet **122**. The magnet **122** may be disposed at an end of the actuator **92***b*. In the open orientation (FIG. **8**A), the end of the actuator **92***b* that includes the magnet **122** may extend outside the first housing **24***a*. For example, the magnet **122** may extend beyond the first side **44** of the first housing **24***a*. As will be explained in more detail below, the magnet **122** may interact with a corresponding feature on the second housing assembly **22***b* to urge movement of the actuator **92***b* (e.g., as the door **14** pivots (e.g., clockwise) relative to the wall **18** from an open orientation (FIG. **8**A) to the closed orientation (FIG. **8**B)).

[0059] The second housing assembly **22***b* may include a second housing **70***b* and the locking system **72**. The locking system **72** may be coupled to the second housing **70***b*. The second housing **70***b* may include a magnet **124** disposed on or within the second housing **70***b*. For example, the magnet **124** may be disposed on or near a first side **126** of the second housing **70***b*. When the door **14** is in the closed orientation (FIG. **8**B), the first side **126** of the second housing **70***b* may face the first housing assembly **20***b*. In some implementations, the magnets **122**, **124** are oriented such that when the door **14** is in the closed orientation (FIG. **8**B), the magnets **122**, **124** have the same poles (e.g., north or south poles) facing one another. In this regard, as the door **14** pivots (e.g., clockwise) from the open orientation (FIG. **8**A) to the closed orientation (FIG. **8**B), the magnetic force between the magnets **122**, **124** increases, thereby urging movement of the actuator **92***b* inside the first housing **24***a*. When the door **14** is in the closed orientation (FIG. **8**B), the magnetic force created between the magnets **122**, **124** may be stronger than the force imparted upon the actuator **92***b* by the one or more biasing members **96**. As the door **14** pivots (e.g., counter-clockwise) from the closed orientation (FIG. **8**B) to the open orientation (FIG. **8**A), the magnetic force between the magnets **122**, **124** decreases. Once the force imparted upon the actuator **92***b* by the one or more biasing members **96** is stronger than the magnetic force, the one or more biasing members **96** may urge the actuator **92***b* to extend outside the first housing **24***a*.

[0060] A method of operating a lock system (e.g., lock systems **10**, **10***a*, **10***b*) will now be described with reference to FIGS. **1**A-**9**. In a step of operating the lock system **10**, the door **14** is moved from the open orientation (FIG. **1**A) to the closed orientation (FIG. **1**B). In the open orientation, the first housing assembly **20**, **20***a*, **20***b* is spaced apart from the second housing assembly **22**, **22***a*, **22***b* and the plug subassembly **26**, **26***a* and the actuator **46**, **46***a* may extend outside of the first housing **24**. For example, the plug subassembly **26**, **26***a* and the actuator **46**, **46***a* may extend beyond the first side **44**.

[0061] As the door **14** moves from the open orientation to the closed orientation, the first housing assembly **20**, **20***a*, **20***b* may contact the second housing assembly **22**, **22***a*, **22***b*. For example, the actuator **46**, **46***a* (e.g., the wheel **50** or the chamfered end **90**) may engage (e.g., roll along or slide against) the ramp **74**. As the actuator **46**, **46***a* moves along the ramp **74**, the ramp **74** may push the actuator **46**, **46***a* into the first housing **24**. The movement of the actuator **46**, **46***a* into the first housing **24** may, in turn, move the actuation system(s) **28** (e.g., the first actuation system **28-1** or the actuation system **28***a*). For example, translation of the actuator **46**, **46***a* towards the first housing **24** may cause the first link **58** to rotate (e.g., counter-clockwise) about the axis A1. Rotation of the first link **58** may cause the second link **60** to rotate about the axis A2. Rotation of the second link **60** may cause the third link **62** to rotate about the axis A5. Rotation of the third link **62** may cause the plug **34**, **34***a* to retract (e.g., translate) inside the first housing **24**. For example, the rotation of the third link **62** may apply a force on the plug **34**, **34***a* that is opposite, and greater than, the force applied by the deadlock assembly **49** (e.g., surface **68**). Translation of the plug **34** within the first housing **24** may cause the link **66** to rotate and allow the plug **34** to retract inside the first housing **24**.

[0062] In configurations with the second actuation system **28-2**, the movement of the second actuation system **28-2** is similar to the movement of the first actuation system **28-1** described above. Movement of the first actuation system **28-1** and the second actuation system **28-2** may occur simultaneously.

[0063] In configurations with the deadlock system **76**, the deadlock system **76** may move from the first orientation (FIG. **3**A) to the second orientation (FIG. **3**B) as the actuation system(s) **28** (e.g., the first actuation system **28-1** or the actuation system **28***a*) moves from the second orientation (FIG. **3**A) to the first orientation (FIG. **3**B). Movement of the deadlock system **76** from the first orientation (FIG. **3**A) to the second orientation (FIG. **3**B) allows the plug subassembly **26**, **26***a* to move from the plugged position (FIGS. **1**A, **3**A) to the open position (FIGS. **1**B, **3**B). Movement of the deadlock system **76** and the actuation system(s) **28** (e.g., the first actuation system **28-1** or the actuation system **28***a*) may occur simultaneously.

[0064] In configurations without an actuation system **28**, **28***a*, the latch **77** contacts the plug subassembly **26**, **26***a* to move from the plugged position (FIGS. **1**A, **3**A, **8**A) to the open position (FIGS. **1**B, **3**B, **8**B).

[0065] When the lock system **10** reaches the closed orientation, the plug subassembly **26** is retracted within the first housing **24** and the latch **77** extends into the opening **42** of the first housing **24**.

[0066] The configuration and operation of the lock systems **10**, **10***a*, **10***b* described herein can prevent an individual (e.g., an inmate) from forcing items (e.g., contraband) into the opening **42** of the first housing **24** when the door **14** is in the open orientation, thus ensuring that the opening can receive the latch **77** when the lock system **10**, **10***a*, **10***b* is in the closed orientation (FIG. **1**B, **3**B, **8**B).

[0067] The following Clauses provide exemplary configurations for a lock system, as described above.

[0068] Clause 1: A lock system comprising: a first housing including a first side defining an opening; a plug disposed within the first housing and configured to move within the opening between an open position and a plugged position; and a first actuation system disposed within the first side of the first housing and configured to move between a first position and a second position, wherein the first actuation system (i) moves the plug from the plugged position to the open position when the first actuation system moves from the first position to the second position, and (ii) secures the plug in the plugged position when the first actuation system moves from the second position to the first position.

[0069] Clause 2: The lock system of clause 1, wherein the plug is disposed within the opening when the plug is in the plugged position.

[0070] Clause 3: The lock system of any of clauses 1 through 2, wherein the first actuation system includes: an actuator; and a linkage assembly coupled to the actuator and the plug.

[0071] Clause 4: The lock system of clause 3, wherein the actuator extends outside the first housing when the first actuation system is in the first position and retracts within the first housing when the first actuation system is in the second position.

[0072] Clause 5: The lock system of clause 4, further comprising a second housing including a latch.

[0073] Clause 6: The lock system of clause 5, wherein the actuator is configured to engage the second housing when the first actuation system is in the second position.

[0074] Clause 7: The lock system of clause 6, wherein the second housing includes a ramp configured to engage the actuator when the first actuation system is in the second position.

[0075] Clause 8: The lock system of clause 7, wherein the first actuation system moves from the first position to the second position as the actuator engages with the ramp.

[0076] Clause 9: The lock system of clause 8, wherein the latch is in a locked position when the plug is in the open position, and the opening is configured to receive the latch when the latch is in the locked position.

[0077] Clause 10: The lock system of any of clauses 1 through 9, further comprising a second actuation system coupled to the plug and configured to move between a third position and a fourth position, wherein the second actuation system moves the plug from the plugged position to the open position when the second actuation system moves from the third position to the fourth position.

[0078] Clause 11: A lock system comprising: a first housing defining an opening; a plug supported by the first housing and configured to move between an open position and a plugged position; and a first actuation system coupled to the plug and configured to move between a first position and a second position, wherein the first actuation system (i) moves the plug from the plugged position to the open position when the first actuation system moves from the first position to the second position, and (ii) secures the plug in the plugged position when the first actuation system moves from the second position to the first position.

[0079] Clause 12: The lock system of clause 11, further comprising a deadlock system coupled to the plug and configured to move between a third position and a fourth position.

[0080] Clause 13: The lock system of clause 12, wherein the deadlock system (i) inhibits movement of the plug in the third position and (ii) allows movement of the plug in the fourth position.

[0081] Clause 14: The lock system of clauses 12 or 13, wherein the deadlock system moves from the third position to the fourth position when the first actuation system moves from the first position to the second position.

[0082] Clause 15: The lock system of any of clauses 12 through 14, wherein the deadlock system includes: an actuator, and a locking lever coupled to the actuator and configured to engage the plug.

[0083] Clause 16: The lock system of clause 15, wherein the plug includes a retaining clip configured to receive the locking lever.

[0084] Clause 17: The lock system of clauses 15 or 16, wherein the plug includes a notch and the locking lever is configured to engage the notch when the deadlock system is in the third position.

[0085] Clause 18: The lock system of any of clauses 15 through 17, wherein the locking lever is disposed above the plug when the deadlock system is in the fourth position.

[0086] Clause 19: The lock system of any of clauses 15 through 18, wherein the actuator includes a chamfered end.

[0087] Clause 20: The lock system of any of clauses 11 through 19, wherein the plug is disposed within the opening when the plug is in the plugged position.

[0088] Clause 21: A lock system comprising: a first housing including a first side defining an opening; a plug disposed within the first housing and configured to move within the opening between an open position and a plugged position; and a deadlock system disposed within the first side of the first housing and coupled to the plug, the deadlock system configured to move between a first position and a second position, wherein the deadlock system (i) inhibits movement of the plug in the first position and (ii) allows movement of the plug in the second position.

[0089] Clause 22: The lock system of clause 21, further comprising a first actuation system coupled to the plug and configured to move between a third position and a fourth position.

[0090] Clause 23: The lock system of clause 22, wherein the first actuation system (i) moves the plug from the plugged position to the open position when the first actuation system moves from the third position to the fourth position, and (ii) secures the plug in the plugged position when the first actuation system moves from the fourth position to the second position.

[0091] Clause 24: The lock system of clauses 22 or 23, wherein the deadlock system moves from the first position to the second position when the first actuation system moves from the third position to the fourth position.

[0092] Clause 25: The lock system of any of clauses 21 through 24, wherein the deadlock system includes: an actuator, and a locking lever rotatably coupled to the actuator and configured to engage the plug.

[0093] Clause 26: The lock system of clause 25, wherein the plug includes a retaining clip configured to engage the locking lever.

[0094] Clause 27: The lock system of clauses 35 or 36, wherein the plug includes a notch configured to receive the locking lever when the deadlock system is in the first position.

[0095] Clause 28: The lock system of any of clauses 35 through 37, wherein the locking lever is configured to rotate relative to the plug when the deadlock system moves between the first position and the second position.

[0096] Clause 29: The lock system of any of clauses 35 through 38, wherein the actuator includes a chamfered end.

[0097] Clause 30: The lock system of any of clauses 31 through 39, wherein the plug is disposed within the opening when the plug is in the plugged position.

CONCLUSION

[0098] The terminology used herein is for the purpose of describing particular exemplary configurations only and is not intended to be limiting. As used herein, the singular articles "a," "an," and "the" may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms "comprises," "comprising," "including," and "having," are inclusive and therefore specify the presence of features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. Additional or alternative steps may be employed.

[0099] When an element or layer is referred to as being "on," "engaged to," "connected to," "attached to," or "coupled to" another element or layer, it may be directly on, engaged, connected, attached, or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being "directly on," "directly engaged to," "directly connected to," "directly attached to," or "directly coupled to" another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., "between" versus "directly between," "adjacent" versus "directly adjacent," etc.). As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items.

[0100] The terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections. These elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as "first," "second," and other numerical terms do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example configurations.

[0101] The term "set" generally means a grouping of one or more elements. The elements of a set do not necessarily need to have any characteristics in common or otherwise belong together. The phrase "at least one of A, B, and C" should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean "at least one of A, at least one of B, and at least one of C." The phrase "at least one of A, B, or C" should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR.

[0102] The foregoing description has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular configuration are generally not limited to that particular configuration, but, where applicable, are interchangeable and can be used in a selected configuration, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

1. A lock system comprising:

a first housing including a first side defining an opening;

a plug disposed within the first housing and configured to move within the opening between an open position and a plugged position; and

a deadlock system disposed within the first side of the first housing and coupled to the plug, the deadlock system configured to move between a first position and a second position,

wherein the deadlock system (i) inhibits movement of the plug in the first position and (ii) allows movement of the plug in the second position.

2. The lock system of claim **1**, further comprising a first actuation system coupled to the plug and configured to move between a third position and a fourth position.

3. The lock system of claim **2**, wherein the first actuation system (i) moves the plug from the plugged position to the open position when the first actuation system moves from the third position to the fourth position, and (ii) secures the plug in the plugged position when the first actuation system moves from the fourth position to the second position.

4. The lock system of claim **3**, wherein the deadlock system moves from the first position to the second position when the first actuation system moves from the third position to the fourth position.

5. The lock system of claim **1**, wherein the deadlock system includes an actuator, and a locking lever rotatably coupled to the actuator and configured to engage the plug.

6. The lock system of claim **5**, wherein the plug includes a retaining clip configured to engage the locking lever.

7. The lock system of claim **5**, wherein the plug includes a notch configured to receive the locking lever when the deadlock system is in the first position.

8. The lock system of claim **5**, wherein the locking lever is configured to rotate relative to the plug when the deadlock system moves between the first position and the second position.

9. The lock system of claim **5**, wherein the actuator includes a chamfered end.

10. The lock system of claim **1**, wherein the plug is disposed within the opening when the plug is in the plugged position.

11. A lock system comprising:

a first housing including a first side defining an opening;

a plug disposed within the first housing and configured to move within the opening between an open position and a plugged position; and

a first actuation system disposed within the first side of the first housing and configured to move between a first position and a second position,

wherein the first actuation system (i) moves the plug from the plugged position to the open position when the first actuation system moves from the first position to the second position, and (ii) secures the plug in the plugged position when the first actuation system moves from the second position to the first position.

12. The lock system of claim **11**, wherein the plug is disposed within the opening when the plug is in the plugged position.

13. The lock system of claim **11**, wherein the first actuation system includes:

an actuator; and

a linkage assembly coupled to the actuator and the plug.

14. The lock system of claim **13**, wherein the actuator extends outside the first housing when the first actuation system is in the first position and retracts within the first housing when the first actuation system is in the second position.

15. The lock system of claim **14**, further comprising a second housing including a latch.

16. The lock system of claim **15**, wherein the actuator is configured to engage the second housing when the first actuation system is in the second position.

17. The lock system of claim **16**, wherein the second housing includes a ramp configured to engage the actuator when the first actuation system is in the second position.

18. The lock system of claim **17**, wherein the first actuation system moves from the first position to the second position as the actuator engages with the ramp.

19. The lock system of claim **18**, wherein the latch is in a locked position when the plug is in the open position, and the opening is configured to receive the latch when the latch is in the locked position.

20. The lock system of claim **11**, further comprising a second actuation system coupled to the plug and configured to move between a third position and a fourth position, wherein the second actuation system automatically moves the plug from the plugged position to the open position when the second actuation system moves from the third position to the fourth position.

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